Difference Between Encapsulation and Abstraction

1. Encapsulation

Encapsulation means wrapping data and methods together in a single unit and protecting the data from direct access.

Example: In Instagram, a user's profile information is protected, and the user can update it through the app's settings.

2. Abstraction

Abstraction means hiding internal working details and showing only the necessary features to the user.

Example: In WhatsApp, we can send messages by typing and clicking the Send button without knowing how the message is transmitted internally.

Key Differences

Encapsulation	Abstraction
It protects data from direct access.	It hides unnecessary implementation details.
It focuses on how data is organized and protected.	It focuses on showing essential features.
Example: Protecting Instagram profile information.	Example: Sending a WhatsApp message without knowing its internal working.

In Simple Words

Encapsulation is about **protecting data**, while abstraction is about **hiding complexity**.